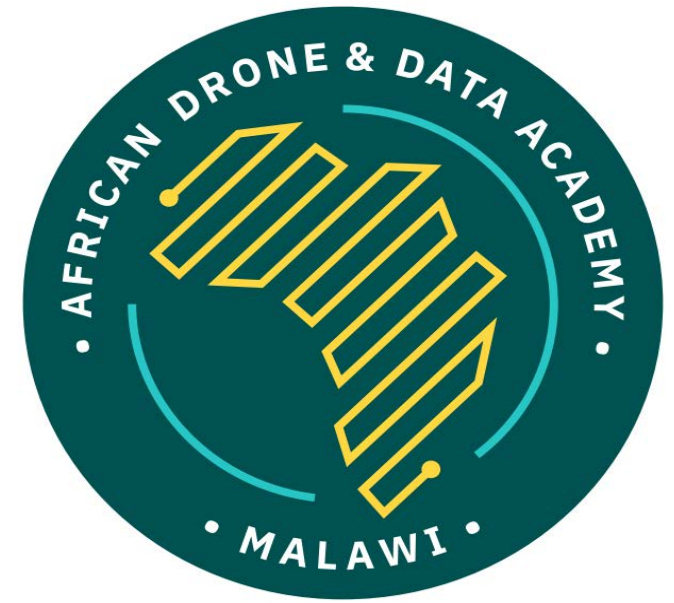




TOP Level 1

Day 1 Version A





TOP Level 1

Principles of Flight Part 1

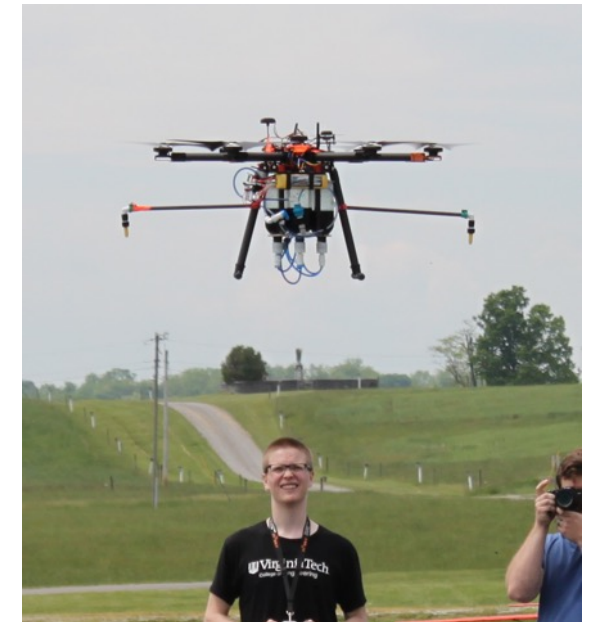


Principles of flight

- In this module you will learn what principles and physics apply to aircraft
- We will look at both fixed wing and VTOL aircraft
- You will gain an understanding of how aircraft are able to convert motion through air into a lifting force

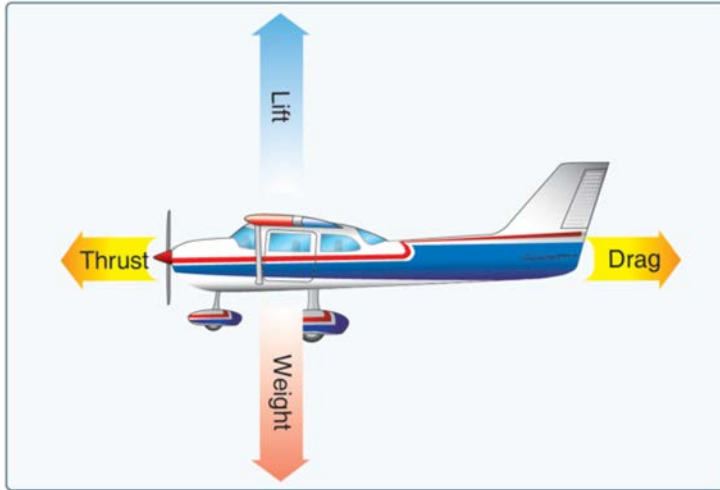


11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

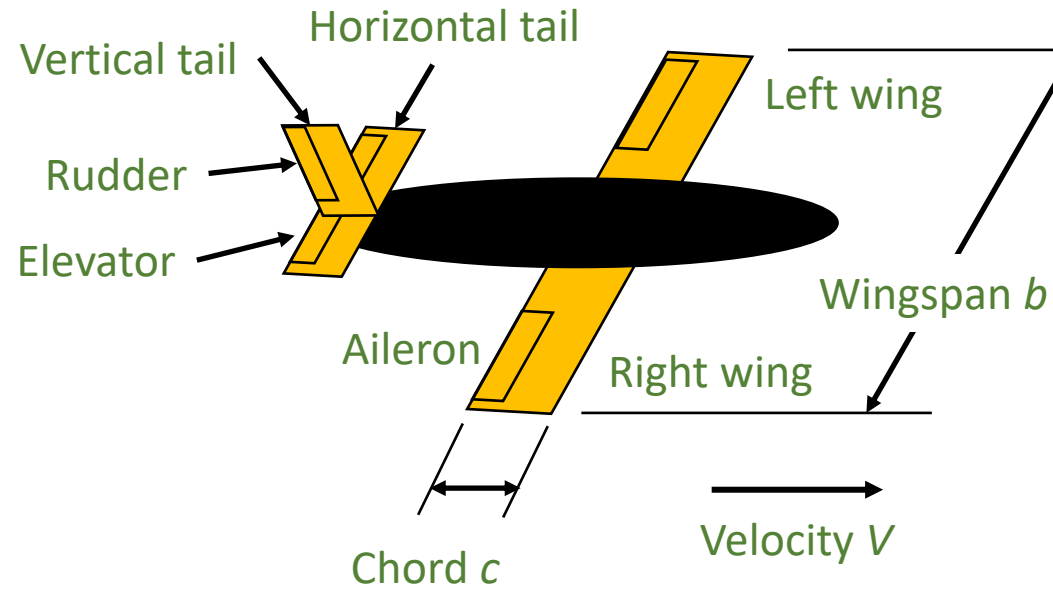


Some useful aircraft definitions

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



The four forces experienced in flight: Lift, Drag, Weight, Thrust



Wing area $S = bc$
Aspect ratio $= b/c$

In the case of a flying wing that does not have a tail, the ailerons also function as the elevator through a mixing of the controls. In this case the surfaces are called *elevons*



How does a wing generate lift?

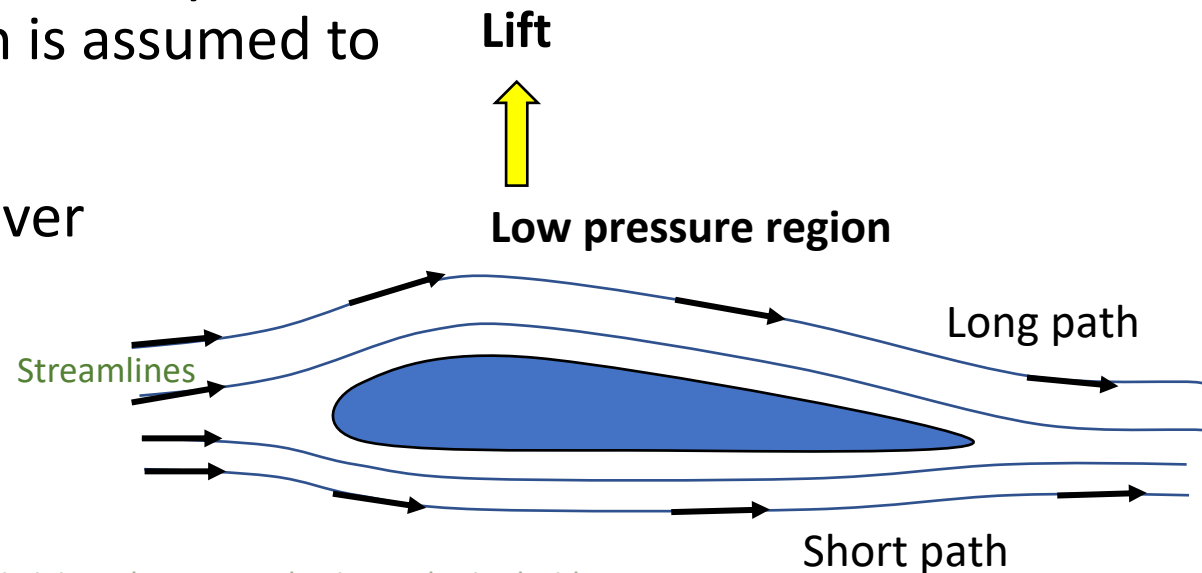
Air is a *fluid* that has *properties* which we can use to create a lifting force

$$\text{Density } \rho = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Pressure } p = \frac{\text{Force}}{\text{Area}}$$

An **airfoil** is a special kind of wing that is only described by its 2D shape - it's length is assumed to be infinite

- Lift is generated when air flowing over the top of the airfoil must travel further than air flowing over the bottom, resulting in a reduction in pressure on the top surface



11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

How does a wing generate lift?

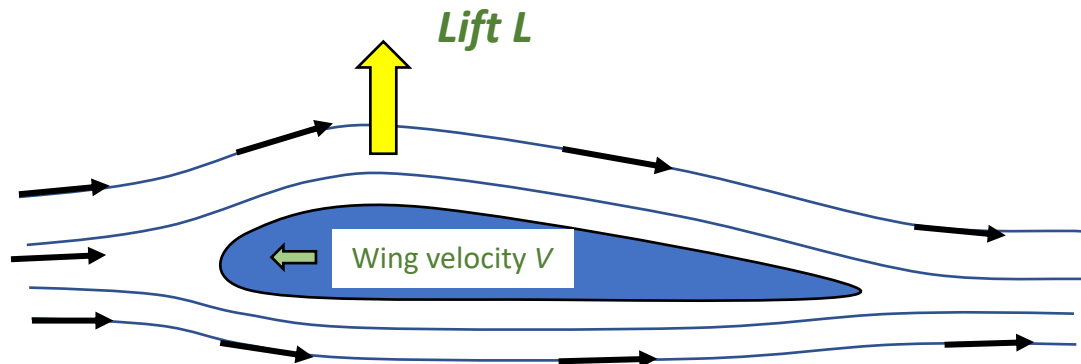
- The change in pressure from atmospheric to the reduced pressure on the top of the wing is described by Bernoulli's equation with C_L and wing velocity V substituted into the equation:

$$p_1 - p_2 = \frac{1}{2}\rho V_2^2 = \frac{1}{2}\rho V^2 C_L$$

- The *lift force* L that is generated by the wing is the difference in pressure multiplied by the wing area:

$$L = S(p_1 - p_2) = \frac{1}{2}\rho V^2 S C_L$$

This is the standard equation describing the lift generated by a wing



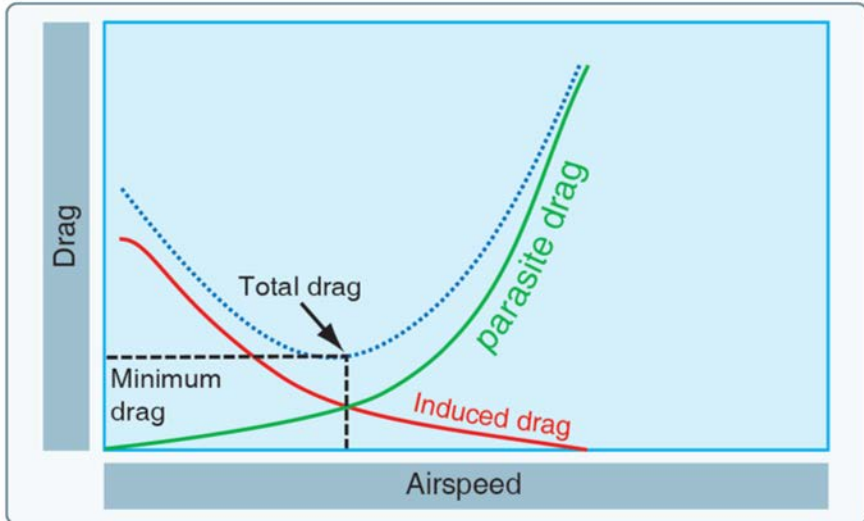
Copyright Virginia Tech: No reproduction authorized without permission

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

The lift coefficient, C_L , is a non-dimensional coefficient used to characterize the lift generated by an airfoil

Lift counteracts the **weight** of the aircraft in level flight while **thrust** is the force that counteracts **drag**

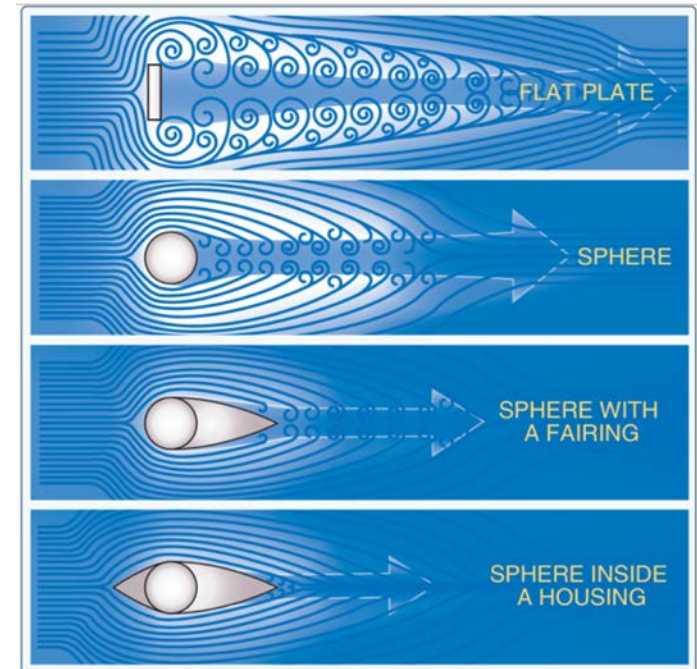
- Drag force is created by the aircraft as it flies through the air
 - The two kinds of drag are *parasite* drag and *induced* drag
 - Parasite drag is caused by the friction the aircraft experiences in flight with the atmosphere
 - Induced drag is drag due to lift generated by the wing



This plot shows that the total drag is minimized when the parasite and induced drag terms are equal

Copyright Virginia Tech: No reproduction authorized without permission

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



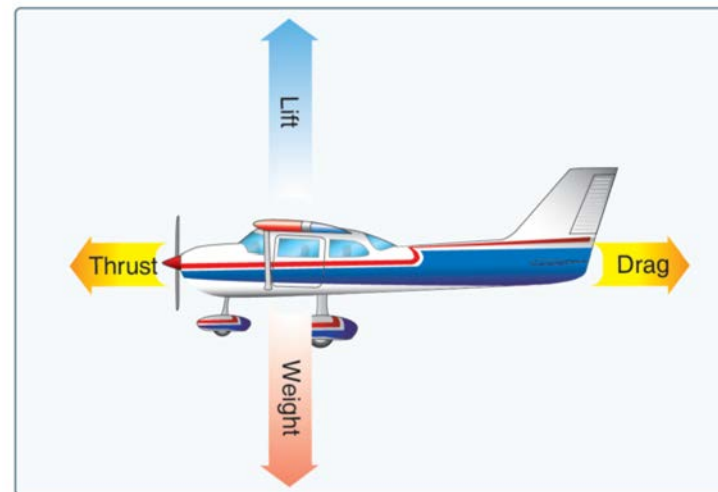
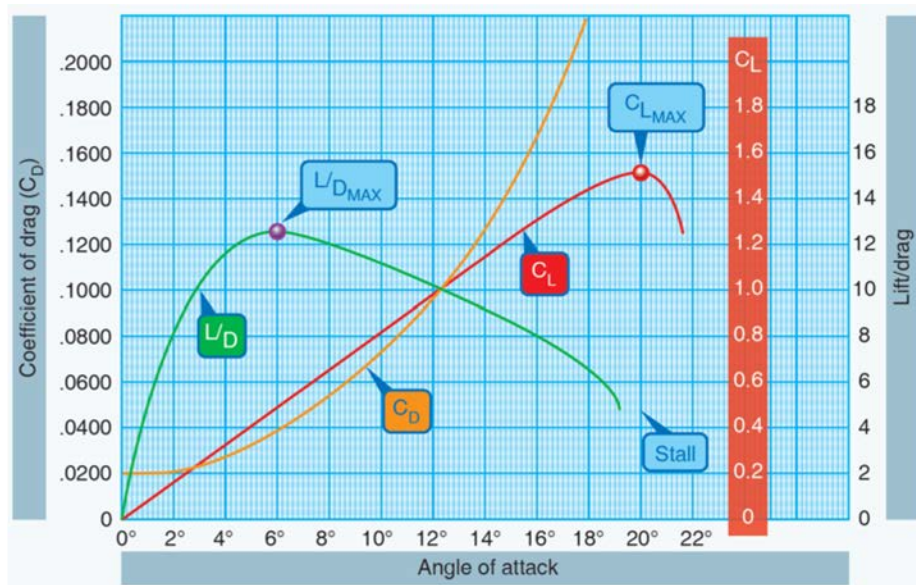
Drag caused by shape

The drag force is calculated just like lift force, only it is applied in the direction of aircraft travel

- The drag coefficient C_D is a non-dimensional coefficient that relates induced drag and parasite drag to the overall drag force on the aircraft
- The drag coefficient C_D is a non-dimensional coefficient that relates induced drag and parasite drag to the overall drag force on the aircraft

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

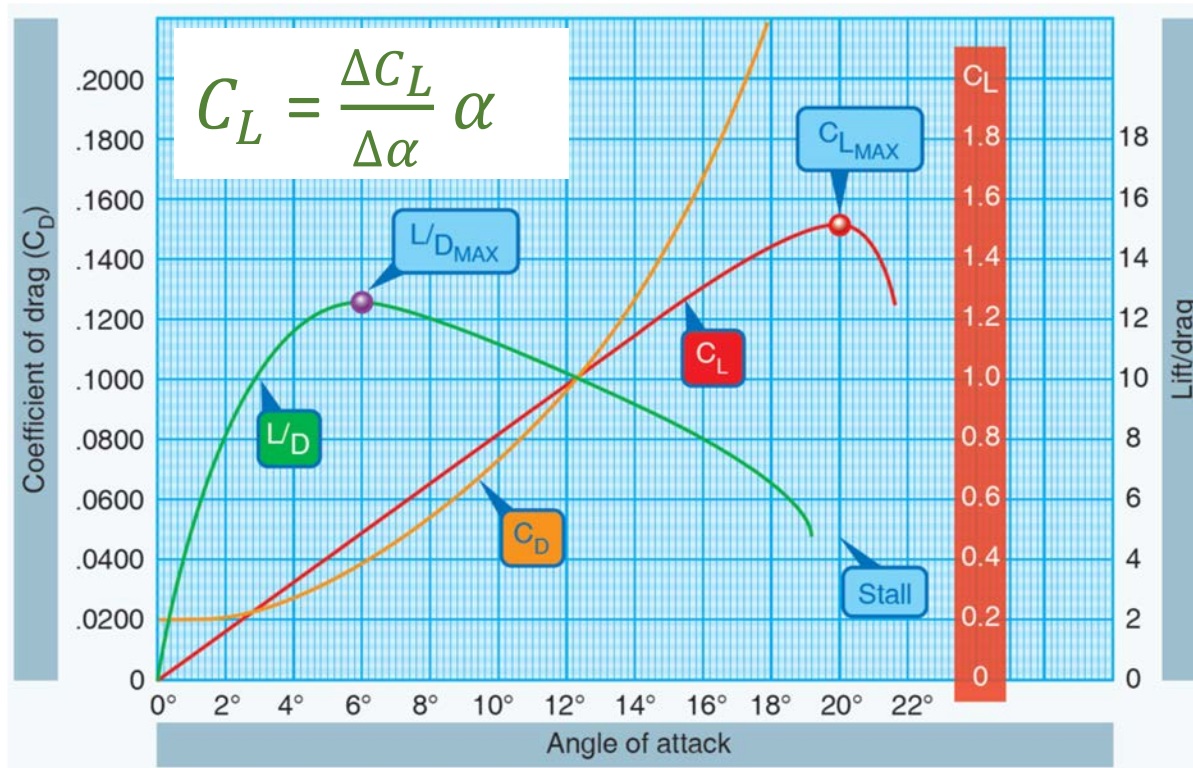
$$D = \frac{1}{2} \rho V^2 S C_D$$



We use wing area S as a reference, but drag is generated by the aircraft fuselage also

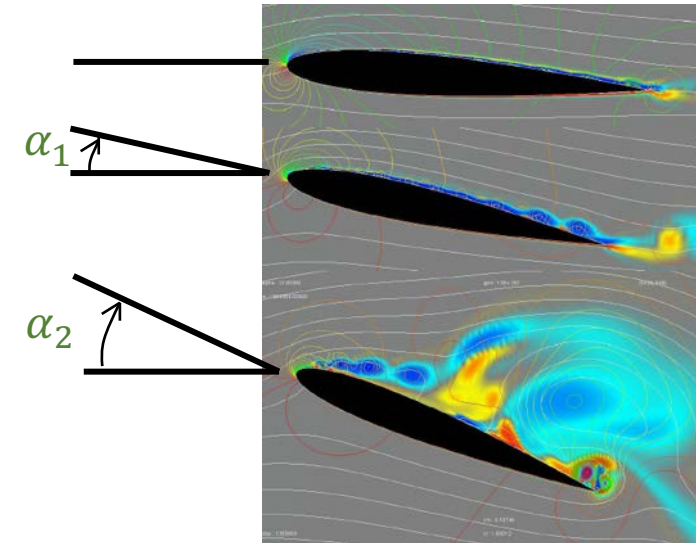
Lift and drag are forces that can be quantified as a function of the angle of attack of an airfoil

- The lift coefficient is a function of angle of attack, α :



Copyright Virginia Tech: No reproduction authorized without permission

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

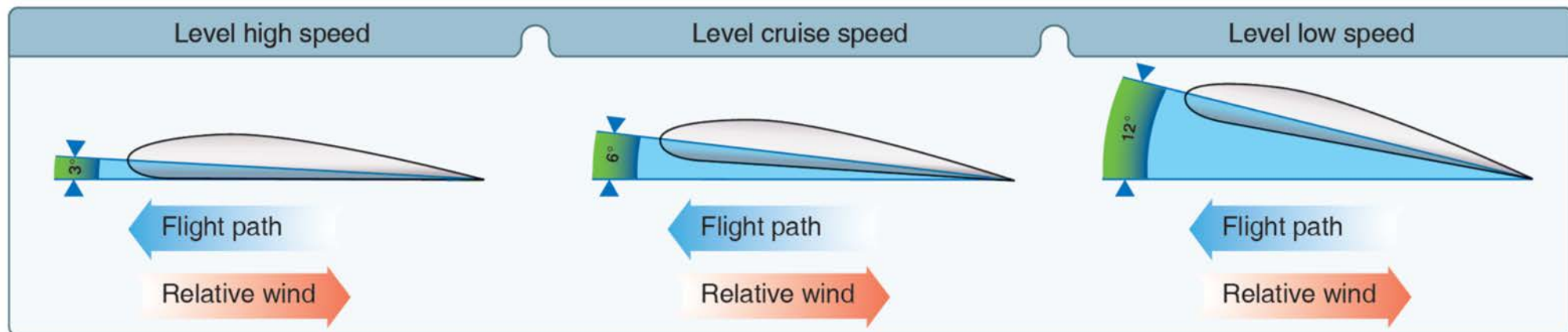


- Drag due to lift is called induced drag
- As the angle of attack increases, both lift and drag increase until the wing stalls

The angle of attack that an aircraft flies at depends on its speed

- A **fast** airplane will fly at a lower angle of attack because V is high and that requires C_L to be lower to maintain the same *Lift* (see equation slide 3)
- A **slow** airplane will fly at a higher angle of attack because V is low and that requires C_L to be higher to maintain the same *Lift* (see equation slide 3)

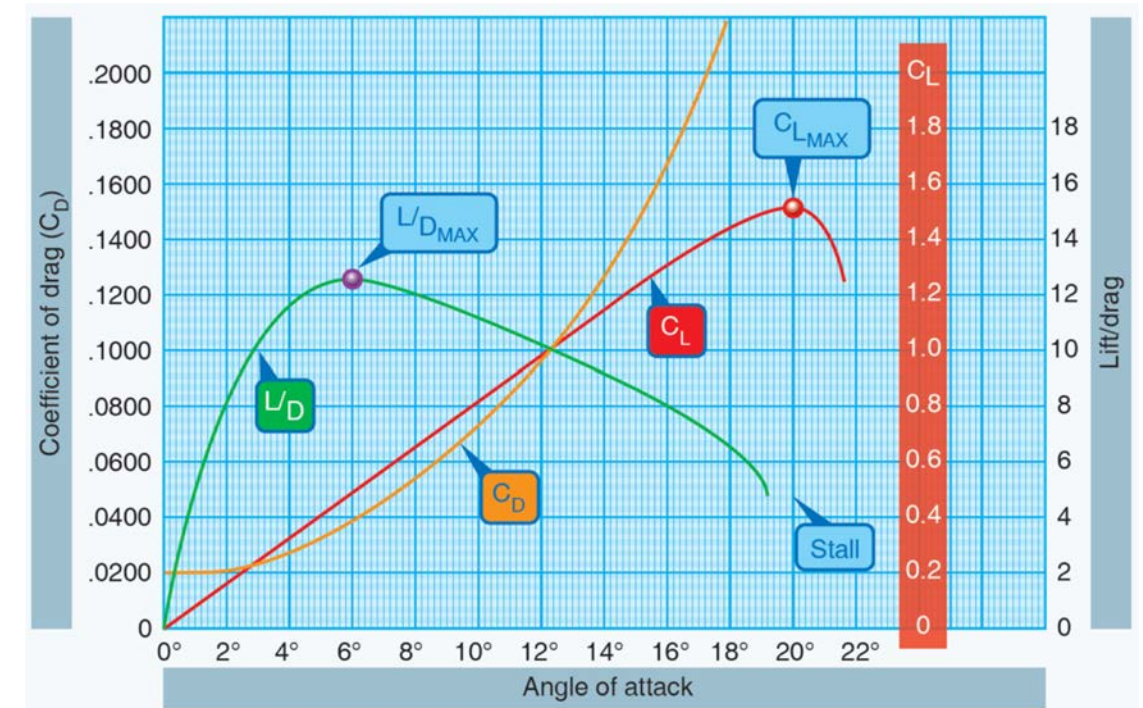
11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. **Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



The stall speed (maximum lift coefficient) should normally be avoided to prevent an uncontrolled descent of the airplane

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

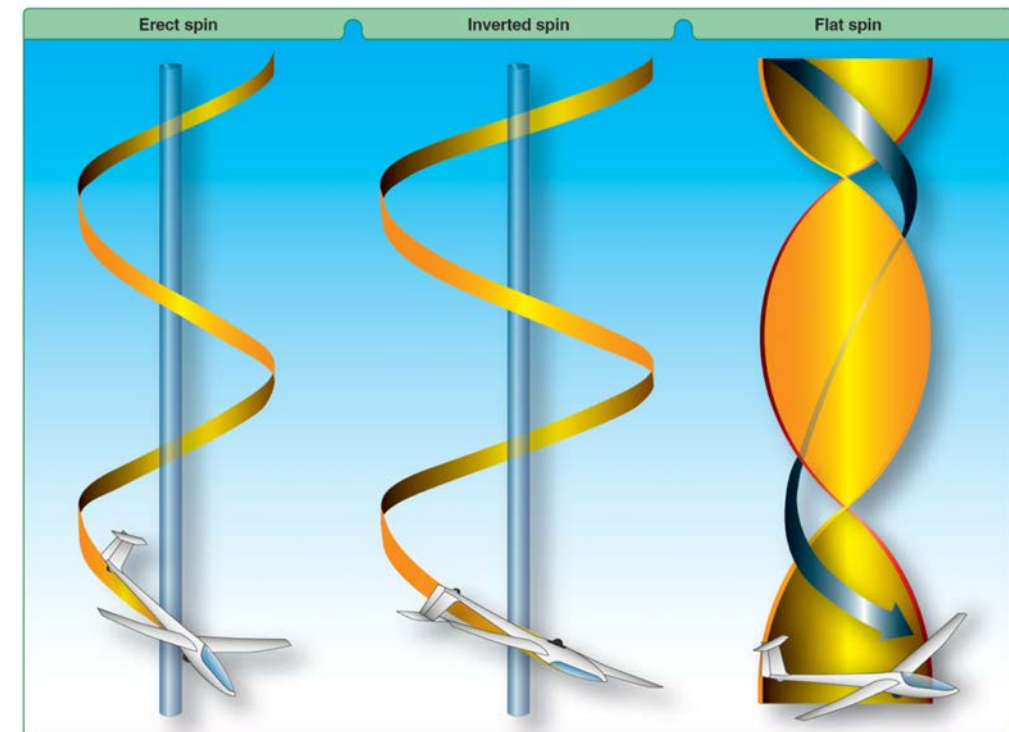
- Airplanes will fly in a lift coefficient range of 0.4 – 1.2
 - What range of angles of attack does this equate to for the given plot (for a given airfoil?)
- This range provides a enough margin away from stall speed that safe flight is possible



The stall speed (maximum lift coefficient) should normally be avoided to prevent an uncontrolled descent of the airplane

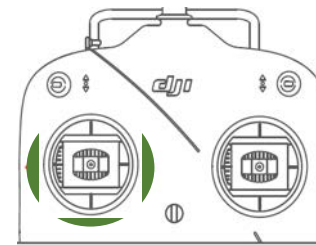
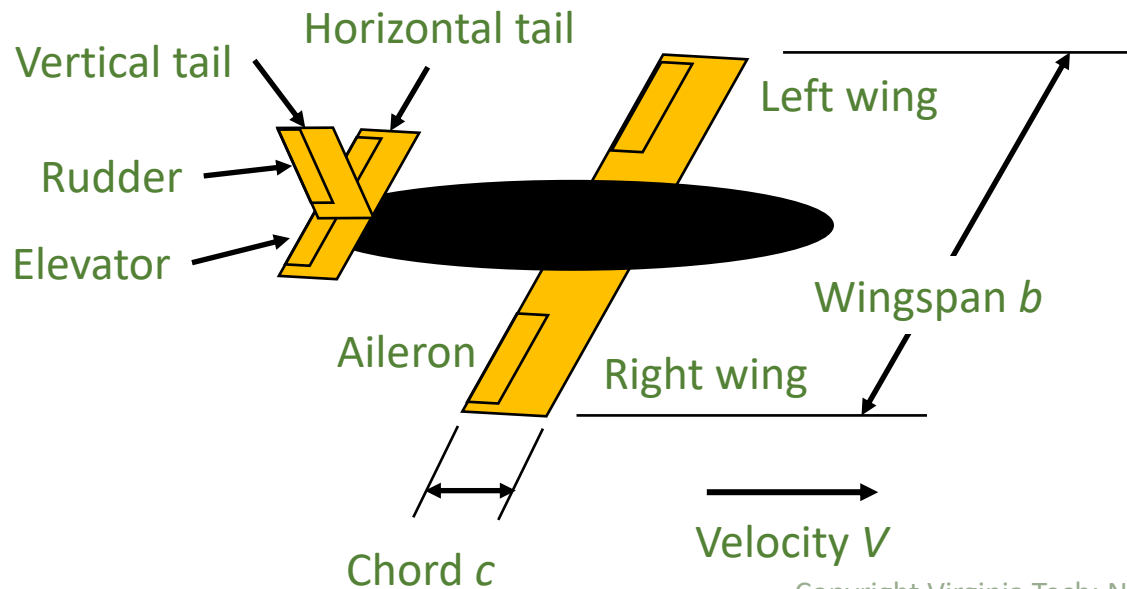
11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

- Airplanes that stall will experience an uncontrolled descent since the control surfaces were not designed to operate in a stalled condition
- The nose of the airplane may drop, or a wing may drop, or the airplane may enter a spin
 - *The pilot of a fixed wing aircraft needs to be aware that the stall speed must be avoided at all times*
 - *Additionally, coordinated flight must be used to prevent the conditions for spin*



Aircraft control is equally important to maintaining flight as is the lift generated by a wing

- **Manual** control of the aircraft is usually described as the pilot's input through the transmitter to actuate the aircraft control surfaces



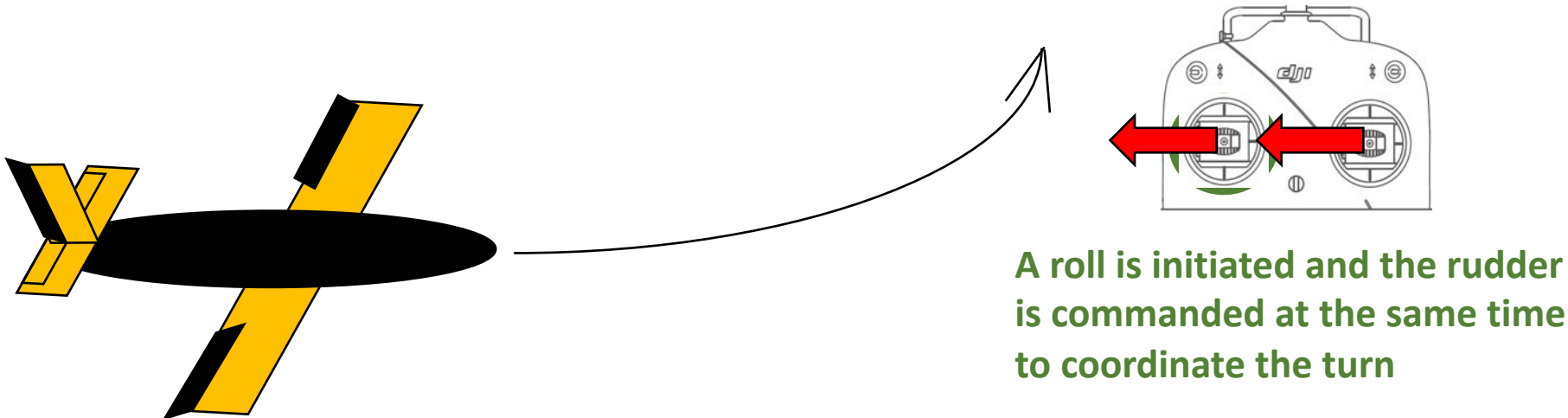
11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



Pilot technique in remote control is similar to manned aircraft control for stall prevention and recovery

- At all times, the aircraft should fly in a **coordinated** manner
 - This means that the rudder is used to trim the nose so that the airflow is aligned with the longitudinal axis of the aircraft

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



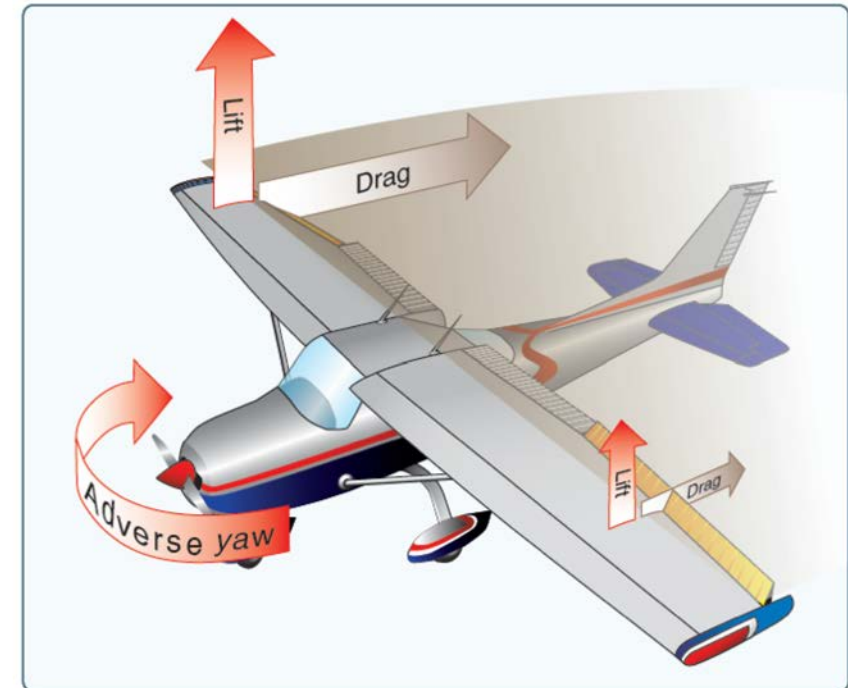
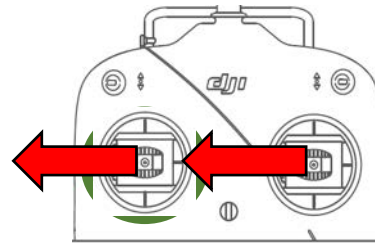
Inputting a rolling command to the right results in an **adverse yaw** to the left - corrected with rudder input

Adverse yaw is the reason that rudder input is required when initiating a turn

- When the ailerons are deflected, **drag is higher** on the aileron which is **deflected downward** because it is generating more lift than upward moving aileron
- The rudder is used to keep the nose from moving in the opposite direction of the turn

To initiate a coordinated turn:

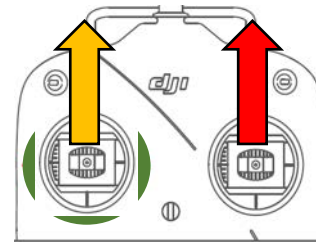
Generally, the rudder is commanded with aileron input initially, then the two surfaces are brought to neutral while the aircraft is slightly trimmed throughout the turn



11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

Stalls and spins

- If the aircraft gets too slow during flight, it has a good chance of stalling and possibly entering a spin
 - *These are dangerous flight conditions that need to be avoided*
- If you notice the nose of the aircraft rising during flight and the speed seems to be decreasing, **lower the nose immediately** to prevent the aircraft losing lift on one or both wings
 - **Increase throttle** if it appears that you have sufficiently executed a recovery

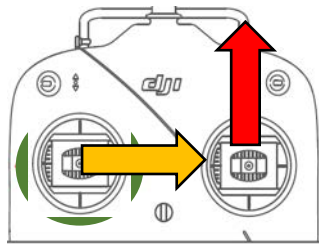


11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

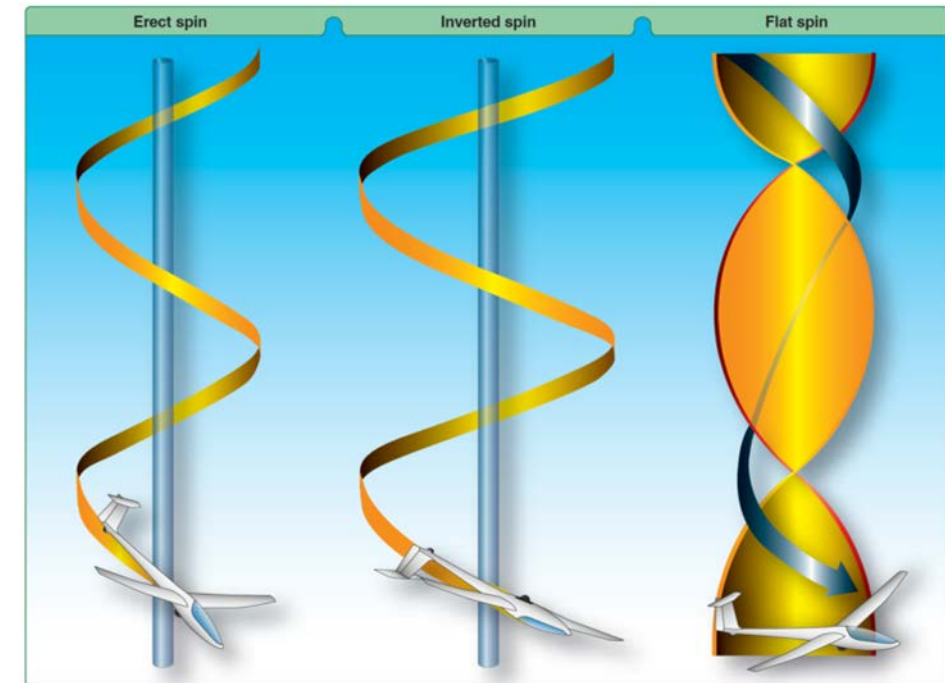


Stalls and spins

- A spin may result if the aircraft has some yawing behavior during the stall
- One wing stalls before the other one, and it drops, pushing the aircraft into a stable turning descent called a spin
- The recovery technique is to drop the nose, and apply opposite rudder to the turn

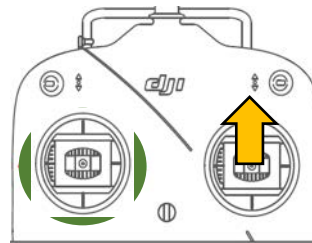


11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



Stalls and spins

- Departure (or tip) stalls and spins are common with small aircraft
- If the aircraft pitches up on takeoff, the likelihood of one wing stalling and causing the plane to roll uncontrollably is good
 - These are very difficult flight regimes to recover from - the aircraft may be too low to recover
 - Flying wing aircraft without a rudder have a limited ability to recover
 - Solution: reduce pitch attitude until sufficient altitude is reached that a recovery may be initiated



11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

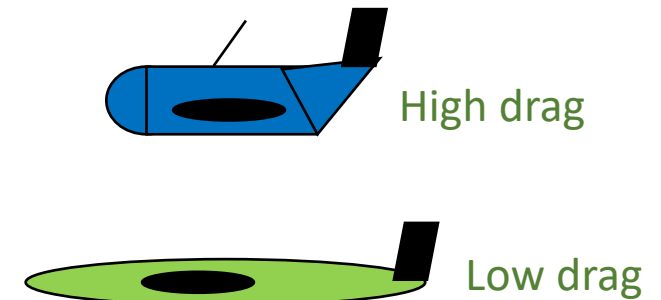


A “tip” stall is caused by the tip of a wing stalling because aerodynamically it may experience higher angles of attack. This can cause the aircraft to suddenly roll out of control

Drag reduction and aerodynamic improvements can improve overall aircraft efficiency

- Fuselage drag cleanup
 - Reduce sharp angles that cause flow separation
 - Enclose antennas inside the fuselage
 - Use fairings that are teardrop shaped
- Wing drag cleanup
 - Make sure control surface hinges do not protrude into the airstream
 - Make sure wing is a smooth, contoured surface with no gaps where connections are made
 - Gaps should be sealed (with tape) to prevent airflow from entering the wing structure

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



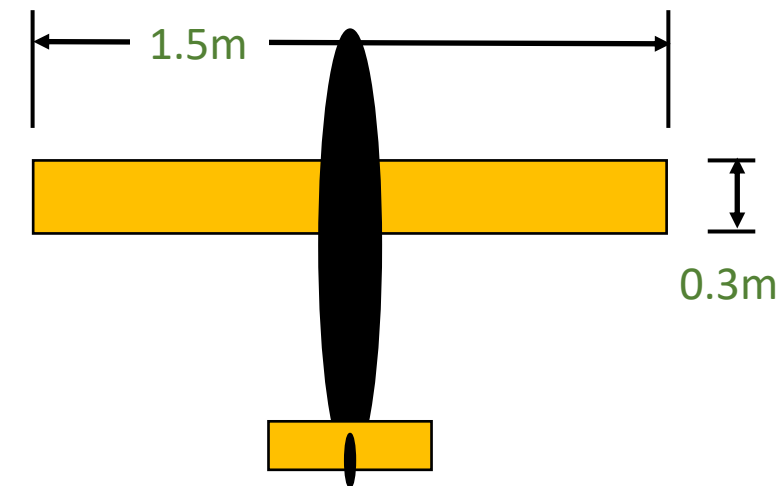
Power consumption in flight

- The power that is consumed by the drone is a direct function of the **drag force** and **speed** of the aircraft
 - The normal definition for power is $P = \text{Force} \times \text{Velocity}$
- In our case, this force is *thrust* from the motor, and is equal to the *drag force* D
- For example, if the drag on an aircraft is 4.85N and it is flying at 20m/s, the power consumed is:

$$P = D \times V = 4.85\text{N} \times 20 \frac{\text{m}}{\text{s}} = 97 \frac{\text{Nm}}{\text{s}} = \mathbf{97W}$$

- Note that this is an *ideal* calculation of power required – in reality, all systems are less than 100% efficient and so an efficiency factor should be used to compute the actual power consumption by the system

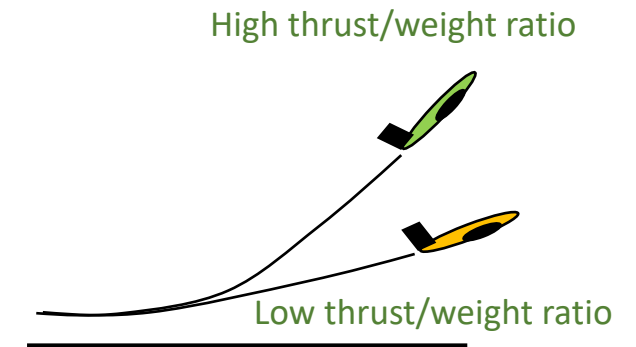
11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



Thrust/weight ratio

- Another useful parameter is the thrust (T) / weight (W) ratio
- The T/W ratio gives an understanding of what performance we may expect from the aircraft
 - A high T/W ratio means the aircraft will climb quickly
 - The penalty is a heavier motor and perhaps more motor drag in cruise that would reduce the range of the aircraft
 - A larger motor may also consume more power in cruise than a smaller motor designed for the cruise power settings it requires

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



The takeoff performance is directly affected by the T/W ratio

Thrust/weight ratio

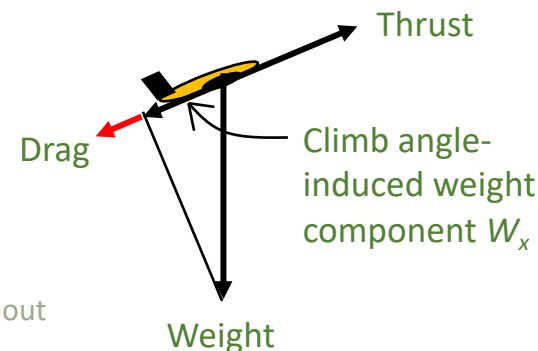
- Using the geometry shown for the climbing aircraft:

$$T = D + W \sin \theta$$

$$\theta = \sin^{-1} \left(\frac{T - D}{W} \right)$$

- Assuming an aircraft has a weight of $10N$, a thrust of $8N$ and a drag force of $2N$, determine the angle of climb:

$$\theta = \sin^{-1} \left(\frac{8 - 2}{10} \right) = 37^\circ$$



11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



Climb angle illustrated

Power/weight ratio

- Another valuable metric to use in evaluating aircraft design is the power/weight ratio
 - Power/weight is a common metric to evaluate the aerodynamic efficiency of the aircraft
 - Metrics used in the general classes of fixed wing and VTOL are useful for comparing performances in those categories

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2

EcoSoar power/weight



$$\begin{aligned} Wt &= 1.25 \text{ kg} \\ P &= 40 \text{ W} \\ \frac{P}{Wt} &= \frac{40}{1250} = 0.032 \text{ W/g} \end{aligned}$$

DJI Mavic Pro



$$\begin{aligned} Wt &= 740 \text{ g} \\ P &= 109 \text{ W} \\ \frac{P}{Wt} &= \frac{109}{740} = 0.147 \text{ W/g} \end{aligned}$$

Power/weight ratio

- The power to fly EcoSoar at cruise speed is **22%** of the power to operate the Mavic 2
- This is typical: fixed wing aircraft are much more efficient than VTOL / rotary wing aircraft which is why their endurance is much longer
- We will look at how VTOL aircraft operate next...

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
- 16. Principles of flight**
17. Human performance
18. Proficiency 1
19. Proficiency 2



$$\frac{P}{Wt} = 0.032 W/g$$



$$\frac{P}{Wt} = 0.147 W/g$$



TOP Level 1

Human Performance



Human performance, threat and error management

- The operation of aircraft is an inherently risky task, and so addressing risk that may occur when human performance is challenged is a critical aspect of safety management
- Section 1.5 of the SMS outlines Occupational Health and Safety guidelines to support crew well being:
 - **Avoid fatigue** by adhering to the following guidelines:
 - **Maximum duty time - 12 hours per day** (this includes any required meetings, standardization flights, ground labs, classes or any other assignment)
 - **Maximum duty time - 70 hours per week**
 - **Minimum of 10 consecutive hours of rest during the 24-hour period** that precedes the completion of the last activity.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2



Human performance, threat and error management

- Section 1.5 of the SMS outlines Occupational Health and Safety guidelines to support crew well being:
 - **Protection from the elements, to include:**
 - Dehydration and heat related risks:
 - Always carry an adequate supply of water and know the warning signs for heat-related illness: **headache, fatigue, cramps, sleepiness, dizziness, low blood pressure, nausea**
 - **Minimize distractions and interference:**
 - Bystanders, spectators and other crew can be a distraction and interference to mission success
 - Use barriers and/or physical markers to reduce the likelihood of non-participant interference in flight operations

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2



Human performance, threat and error management

- **Five traits of pilots that indicate a higher likelihood of accidents occurring (FAA 8083-25B):**

1. Disdain for rules
2. A record of accidents in motor vehicle operations
3. “Thrill and adventure seeking” personality
4. Impulsive rather than methodical and disciplined, both in information gathering and the speed and selection of actions taken
5. Have a disregard for, or tend to under-utilize outside sources of information

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2



External pressures can play a significant role in human performance

- **External pressures** are influences external to the flight that impact the pilot's decision making abilities, including:
 - An over-achieving desire to demonstrate competency
 - A desire to satisfy a specific personal goal (complete five flights in 10 minutes)
 - Emphasis on goal-oriented achievement
 - Emotional pressure, such as pride
- A “personal standard operating procedure (SOP)” can be used to control external pressures:
 - Build extra time into any process
 - Have alternate plans (plan B) if plan A does not work
 - Manage expectations of those who are creating the sense of pressure

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2



Work safety involves creating a safe and productive work environment for everyone involved

- **Know the warning signs of physical distress that may impact your ability to perform (IMSAFE checklist):**

I *Illness*, Do I have any symptoms?

M *Medication*, Have I been taking any prescription or over-the-counter drugs that might affect my performance?

S *Stress*, Am I under psychological pressure from the job? Am I worried about financial matters, health problems, or family discord?

A *Alcohol*, Have I been drinking within the past 8 hours?

F *Fatigue*, Am I tired and not adequately rested?

E *Emotion*, Am I emotionally upset?

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
17. **Human performance**
18. Proficiency 1
19. Proficiency 2



The “CARE” checklist is a way to process perceived hazards in a logical manner

- For each hazard identified, the pilot can use the CARE checklist to address all influencing factors
 - **Consequences:** What (bad) will happen if the proposed action is taken?
 - **Alternatives:** What alternative actions may be used to mitigate a negative outcome?
 - **Reality:** Come to terms with the risks that are present and how they need to be mitigated
 - **External factors:** Acknowledge that external factors will influence your decision making

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2



Following up with CARE, the “TEAM” checklist can be used to choose and implement risk controls

- TEAM provides a process for decision-making to move forward with confidence
 - **Transfer:** Should the risk decision be transferred to someone else (head of training)?
 - **Eliminate:** Can the hazard be directly eliminated?
 - **Accept:** Do the benefits of accepting the risk outweigh the costs?
 - **Mitigate:** What can you do to mitigate the risk?
- Sometimes, the management of risk is a progressive/dynamic process that must be continually evaluated during an operation
 - CARE and TEAM can be referenced throughout a flight for successful operational outcomes

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
15. Commercial ops
16. Principles of flight
- 17. Human performance**
18. Proficiency 1
19. Proficiency 2

